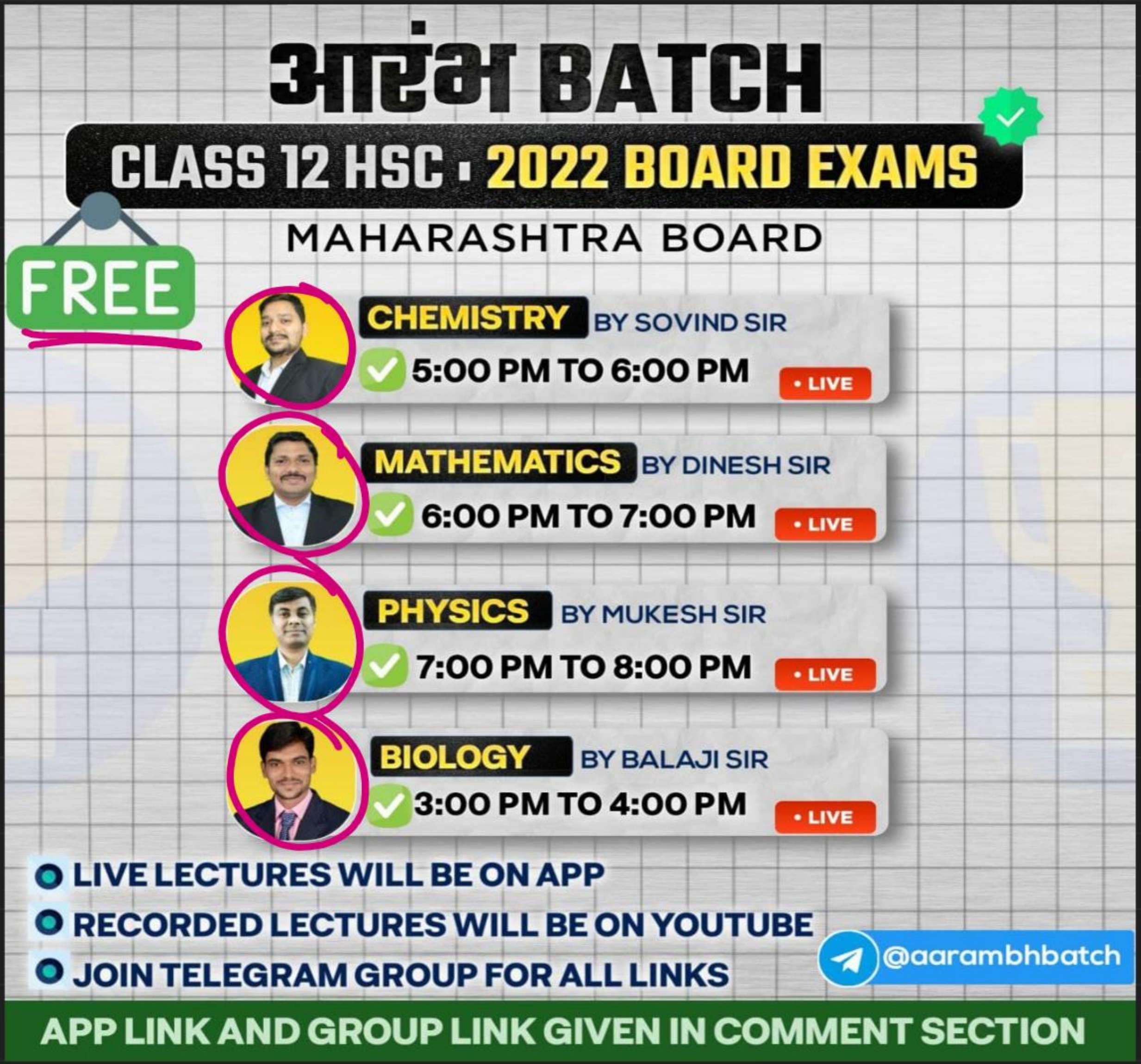

CLASS 11
MAHARASHTRA BOARD
MATHS I
CHAPTER NO. 9
PROBABILITY
1
EXERCISE 9.1
BY DINESH SIR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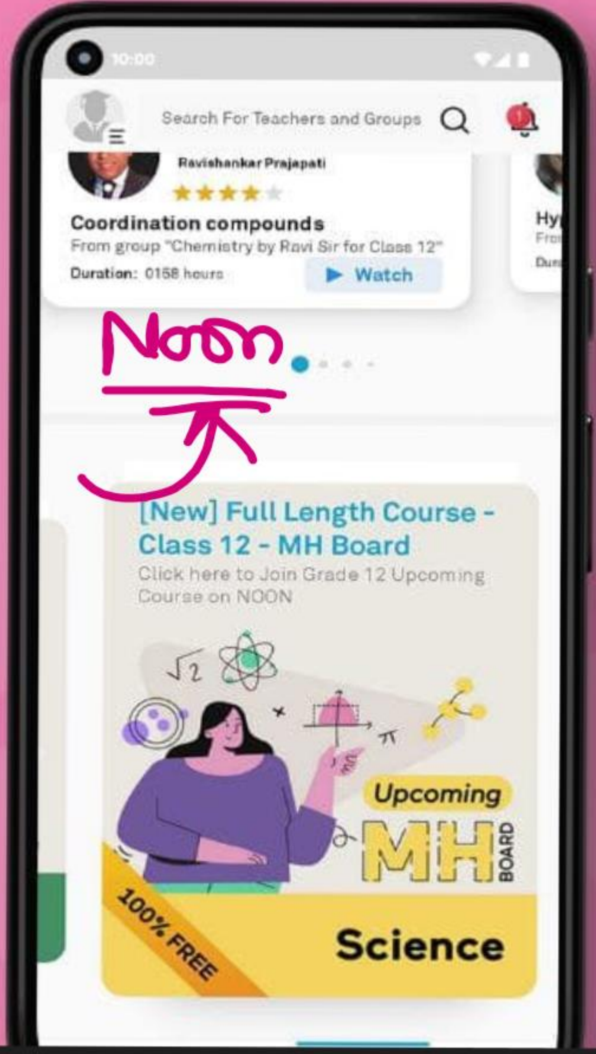

Noon

9 Probability

संभाव्यता

probability

chance

Let's Recall

9.1.1 Basic Terminologies

Random Experiment : Suppose an experiment having more than one outcome. All possible results are known but the actual result cannot be predicted such an experiment is called a random experiment.

Outcome: A possible result of random experiment is called a possible outcome of the experiment.

Sample space: The set of all possible outcomes of a random experiment is called the sample space. The sample space is denoted by S or Greek letter omega (Ω). The number of elements in S is denoted by n(S). A possible outcome is also called a sample point since it is an element in the sample space.

Event: A subset of the sample space is called an event.

Favourable Outcome: An outcome that belongs to the specified event is called a favourable outcome.

Sample space

$S = \{H, T\}$

$S = \{HH, HT, TH, TT\}$

Event

odd

$\boxed{S = \{1, 2, 3, 4, 5, 6\}}$

$A = \{1, 3, 5\}$

$B = \{2, 3, 5\}$

$C = \{1, 3, 5\}$

$D = \{1, 4\}$

Types of Events:

Elementary Event: An event consisting of a single outcome is called an elementary event.

Certain Event: The sample space is called the certain event if all possible outcomes are favourable outcomes. i.e. the event consists of the whole sample space.

Impossible Event: The empty set is called impossible event as no possible outcome is favorable.

$A = \{1\}$

$B = \{3\}$

Algebra of Events:

Events are subsets of the sample space. Algebra of events uses operations in set theory to define new events in terms of known events.

Union of Two Events: Let A and B be two events in the sample space S. The union of A and B is denoted by $A \cup B$ and is the set of all possible outcomes that belong to at least one of A and B.

$A = \{1, 2, \underline{3}, \underline{4}, 5\}$

$B = \{\underline{3}, \underline{4}, 6\}$

$\rightarrow A \cup B = \{1, 2, 3, 4, 5, 6\}$

$\rightarrow A \cap B = \{3, 4\}$

Exhaustive Events: Two events A and B in the sample space S are said to be exhaustive if $A \cup B = S$.

Intersection of Two Events: Let A and B be two events in the sample space S. The intersection of A and B is the event consisting of outcomes that belong to both the events A and B.

Mutually Exclusive Events: Event A and B in the sample space S are said to be mutually exclusive if they have no outcomes in common. In other words, the intersection of mutually exclusive events is empty. Mutually exclusive events are also called disjoint events.

$S = \{HH, HT, TH, TT\}$ $\quad n(S) = 4$

$A = \{HH, HT, TH\}$ $\quad n(A) = 3$

$$P(A) = \frac{n(A)}{n(S)}$$

$$\boxed{P(A) = \frac{3}{4}}$$

EXERCISE 9.1

1) There are four pens: Red, Green, Blue and Purple in a desk drawer of which two pens are selected at random one after the other with replacement. State the sample space and the following events.

a) A : Selecting at least one red pen.

b) B : Two pens of the same color are not selected.

$S = \{RG, RB, RP, RR, GR, GB, GP, GG, BR, BG, BP, BB, PR, PG, PB, PP\}$

$\boxed{n(S) = 16}$

$A = \{RG, RB, RP, RR, GR, BR, PR\}$

$n(A) = 7$ //

$B = \{RG, RB, RP, GR, GB, GP, BR, BG, BP, PR, PG, PB\}$

$n(B) = 12$ //

EXERCISE 9.1

2) A coin and a die are tossed simultaneously. Enumerate the sample space and the following events.

a) A : Getting a Tail and an Odd number

b) B : Getting a prime number

c) C : Getting a head and a perfect square.

When a coin and a die are tossed,

$S = \{H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6\}$

2, 3, 5

$\boxed{n(S) = 12}$

Event A – Getting a tail and odd no.

$A = \{T1, T3, T5\}$ $\boxed{n(A) = 3}$

Even B – Getting a prime number

$B = \{H2, H3, H5, T2, T3, T5\}$ $\boxed{n(B) = 6}$

Event C – Getting a head & perfect square

$C = \{H1, H4\}$ $\boxed{n(C) = 2}$

Hello

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2

EXERCISE 9.1

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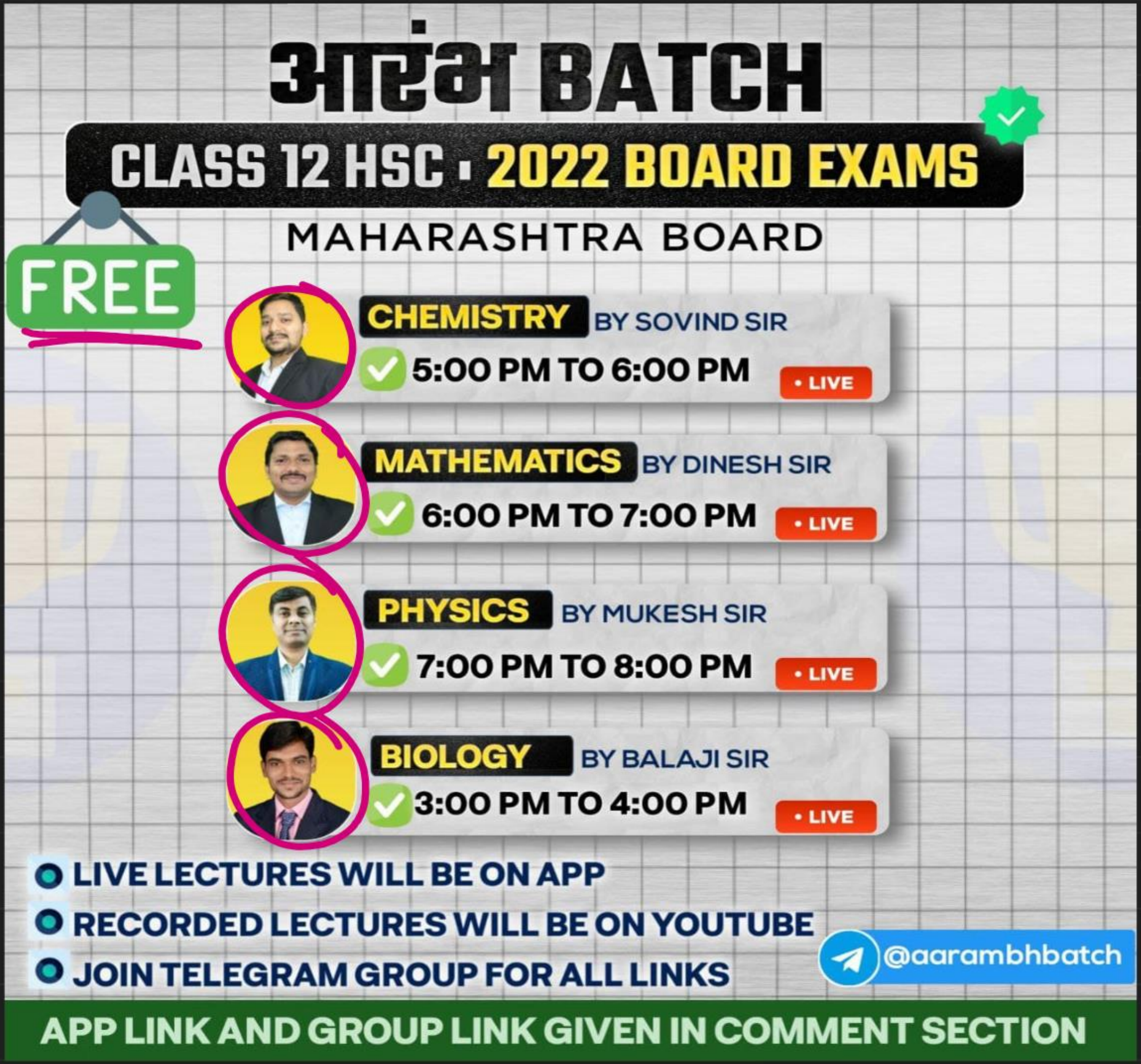
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3) Find n(S) for each of the following random experiments.

a) From an urn containing 5 gold and 3 silver coins, $\boxed{\text{3 coins}}$ are drawn at random

Sol^n Total no. of coins $5+3=8$ coins

$$n(S) = {}^{8}C_{3} = \frac{8\times 7\times \cancel{6}}{\cancel{3\times 2}\times 1} = 56$$

3) Find n(S) for each of the following random experiments.

b) 5 letters are to be placed into 5 envelopes such that no envelop is empty.

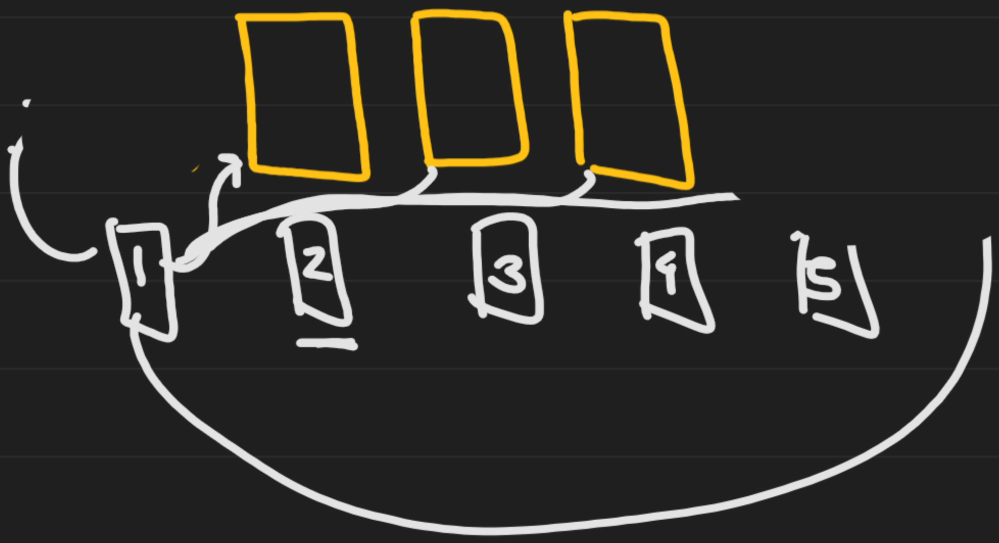
1st letter can be placed in envelop in 5 ways

2nd letter — 4 ways

3rd letter — 3 ways

4th letter — 2 ways

5th letter — 1 ways



$$n(S) = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

3) Find $\boxed{n(S)}$ for each of the following random experiments.

c) 6 books of different subjects arranged on a shelf.

6 books can be arranged among themselves in ${}^{6}P_{6}$ ways

$$n(S) = {}^{6}P_{6} = 6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1$$
$$= 720$$

3) Find $\boxed{n(S)}$ for each of the following random experiments.

d) $\boxed{3 \text{ tickets}}$ are drawn from a box containing $\boxed{20 \text{ lottery}}$ tickets.

3 tickes from 20 tickets can be drawn in ${}^{20}C_3$ ways

$$n(S) = {}^{20}C_3 = \frac{20 \times 19 \times \cancel{18}^{\cancel{6}^{3}}}{\cancel{3} \times \cancel{2} \times 1} = 1140$$

$n(S) = 1140$

4) Two fair dice are thrown. State the sample space and write the favorable outcomes for the following events.

a) A : Sum of numbers on two dice is divisible by 3 or 4.

$$S = \{ \begin{array}{cccccc} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6) \\ (2,1) & \boxed{(2,2)} & (2,3) & (2,4) & (2,5) & (2,6) \\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) & (3,6) \\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & (4,6) \\ (5,1) & (5,2) & (5,3) & \boxed{(5,4)} & (5,5) & (5,6) \\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6) \end{array} \}$$

1, 3, 5

2, 4, 6

3	4
6	8
9	12
12	

$n(S) = 36$

Event A : Sum of nos. divisible by 3 or 4

$$A = \{ (2,1) (1,2) (3,1) (2,2) (1,3) \\ (5,1) (4,2) (3,3) (2,4) (1,5) \\ (6,2) (5,3) (4,4) (3,5) (2,6) \\ (6,3) (5,4) (4,5) (3,6) (6,6) \}$$

$n(A) = 20$

4) Two fair dice are thrown. State the sample space and write the favorable outcomes for the following events.

b) B : Sum of numbers on two dice is 7.

$B = \{(6,1)(5,2)(4,3)(3,4)(2,5)(1,6)\}$

$\boxed{n(B) = 6}$

$\cancel{1}, 2, \cancel{3}, \cancel{4}, 5, \cancel{6}, \cancel{7}, \cancel{8}, \cancel{9}, 10, 11, \cancel{12}$

$A \cup B \neq S$

4) Two fair dice are thrown. State the sample space and write the favorable outcomes for the following events.

4) T
spa
the

c) C : Odd number on the first die. 1,

$$C = \{(1,1)\ (1,2)\ (1,3)\ (1,4)\ (1,5)\ (1,6)$$
$$(3,1)\ (3,2)\ (3,3)\ (3,4)\ (3,5)\ (3,6)$$
$$(5,1)\ (5,2)\ (5,3)\ (5,4)\ (5,5)\ (5,6)\}$$

$$\boxed{n(C) = 18}$$

4) Two fair dice are thrown. State the sample space and write the favorable outcomes for the following events.

d) D : Even number on the first die.

$$D = \{ (2,1)(2,2)(2,3)(2,4)(2,5)(2,6)$$
$$(4,1)(4,2)(4,3)(4,4)(4,5)(4,6)$$
$$(6,1)(6,2)(6,3)(6,4)(6,5)(6,6) \}$$

$$\boxed{n(D) = 18}$$

4) Two fair dice are thrown. State the sample space and write the favorable outcomes for the following events.

e) Check whether events A and B are mutually exclusive and exhaustive.

$A \cap B = \phi$ A and B are mutually exclusive

$A \cup B \neq S$

$\therefore$ A and B are not mutually exhaustive

4) Two fair dice are thrown. State the sample space and write the favorable outcomes for the following events.

f) Check whether events C and D are mutually exclusive and exhaustive.

$C \cap D = \phi$ C & D are mutually exclusive events

$C \cup D = S$ C & D are mutually exhaustive events.